BEAM DESIGN SUMMARY

Project Name	:	G+2 RB BEAM DESIGN @SMR
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-13.625 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	13.625 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	13.625 m	:	177	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	6.31	7.5	6.19	11.36	2.27	14.13
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	13.19	8.59	14.17
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No	:	**B2**	

Group No		:	**G2**	
Analysis Reference(Member)	13.625 m	:	179	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	5.01	8.43	5.87	9.77	0.6	9.02
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	15.65	8.09	13.78
Asv Torsion (sqmm/m)	95.1	47.4	79.72
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No		:	**B3**	
Group No		:	**G3**	
Analysis Reference(Member)	13.625 m	:	180	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm

Design Code	:	IS 456 : 2000 + IS 13920 : 2016
Beam Type	:	Regular Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	9.5	7.57	1.68	0.89	3.63	10.03
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	8.49	13.17	16
Asv Torsion (sqmm/m)	55.79	78.56	101.15
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No		**:**	**B4**	
Group No		**:**	**G4**	
Analysis Reference(Member)	13.625 m	:	178	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.1	10.03	0.8	0.8	6.66	16.95
PtClc (%)	0.228	0.2	0.2	0.2	0.2	0.241
AstCalc (sqmm)	136.27	119.6	119.6	119.6	119.6	143.91
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	12.76	18.22	19.98
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -